

a much wider field of view and a much higher resolution in spite of being much lighter than any head mounted VR device till then. The Rift was born.

From 2009 to 2012, Luckey spent much of time to create a working prototype of his vision. Throughout this duration, he shared his idea and progress on an online VR-enthusiast's forum, where, instead of being stolen by someone else, it was spotted by John Carmack, the legendary video game programmer of Doom, Quake and Wolfenstein-3D fame. When he received a prototype and used it, he was so blown away by the experience that he repurposed his game "Doom 3" to work with the Rift and demoed it at the 2012 June E3 Convention. The rest, we might say, is virtual history.

Over the next two years, Luckey founded Oculus VR with his partners and a Kickstarter that netted \$2.4million – about 1000% of the initial goal – and caught the attention of Facebook founder Mark Zuckerberg. After a demo at the Oculus offices in California, in march 2014 Facebook decided to acquire Oculus VR for \$2billion in cash and shares. After two development kits and some prototypes, the consumer version of the Oculus Rift will be shipping in the first quarter of 2016 with pre-orders having started on 06-Jan, although not to India and we still do not know when that is going to happen.

The Hardware

The Rift is not a standalone device and has to be connected to a PC running Windows which will handle the computational share of the VR experience. Hence, a powerful GPU equivalent to the NVIDIA GeForce GTX 970 or the AMD R9 290, and a powerful CPU equivalent of Intel i5-4590 are needed to handle the Rift's graphics.

The headset itself uses 1080x1200 resolution OLED panels, one for each eye. A combination of high refresh rate (90Hz), global refresh and low persistence (less than 2 milliseconds per frame) allows the experience to be free of any motion blur or nausea of conventional screens. Various dials and additional interfaces allow a variety of facial structures, usage with glasses and accommodates for different inter-pupillary distances as well.

On the audio side, the headset comes with integrated headphones developed with RealSpace 3D Audio technology. Full 6 degree position and rotational tracking is done by the Rift's tracking system, Constellation. Using a series of LED's placed on the device blinking in specific configuration and externally placed infrared, this system provides sub millimeter accuracy with high precision and low latency.